Midterm review

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Midterm #1

- Date
  - ✓ 9/25 (Thursday) during the class
- Closed book
  - ✓ One cheat paper is allowed (for both sides)
  - ✓ You can make the cheat paper either printed or by handwritten.
  - ✓ But, NO calculator, NO electric devices

- 5 questions, each has subproblems
  - ✓ Covers all topics from lecture slides
  - ✓ Complete the pseudocode
  - ✓ Understand the divide-and-conquer approach
  - ✓ Find the solutions from a recurrence equation
    - Exact solutions
    - Asymptotic solutions

Example 1

- For the given Algorithm function,
  1) Determine the total number of iterations executed by the loop structure, i.e., the number of iterations of L4.
  2) Compute the final value of s after Algorithm completes.

```
Algorithm Loop (n):
L1. s = 0
L2. For i = 1 to n^2 do
L3.         For j = 1 to i do
L4.                 s = s + i
```

Example 2 – 1)

- For the given recurrence equation.

$$T(n) = \begin{cases} 1 & , if\ n = 1 \\ 2T\left(\frac{n}{2}\right) + n & , if\ n > 1 \end{cases}$$

1) What is asymptotic solution of $T(n)$?

Example 2 – 2)

2) Find an exact solution for $T(n)$ when n is a power of 2. (Try to solve it by math without using a recursion tree).

Example 2 – 3)

3) Demonstrate the correctness of your result of 2) by showing $T(16)$

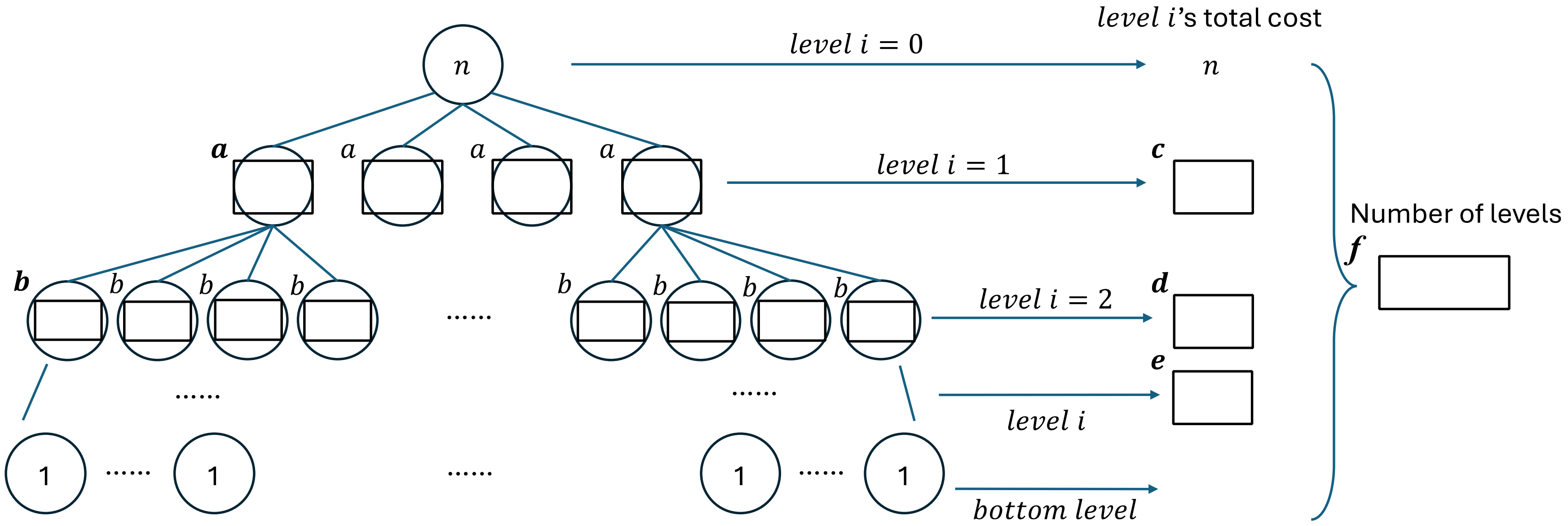
*Hint: find the value of $T(16)$ using recurrence equation, then check with the answer from 2).

Example 3 – 1)

- For the given recurrence equation.

$$T(n) = \begin{cases} 1 & , if\ n = 0 \\ 4T\left(\frac{n}{2}\right) + n & , if\ n > 1 \end{cases}$$

1) Fill out the blanks (a to f) while sketching the recursion tree of the above recurrence.



Example 3 – 2)

2) Now, find the exact solution of the above recurrence based on the results of 1).

*Hint: You may use the following geometric series: $$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}$$

Example 4 – 1)

- For the given array A,

$$A = [-10,\ 2,\ 5,\ -3,\ 14,\ 6,\ 2,\ -5]$$

1) Find the maximum sum of any contiguous subarray in array A. (Justify your answer)

Example 4 – 2)

- ✓ Find that contiguous subarray of A that results the maximum sum, i.e., your answer (subarray) should give the maximum sum of the above question.

Example 5

- Complete the following recursive algorithm FIND_MIN_MAX, employing a divide-and-conquer approach, to find both the minimum and the maximum elements in an array (A) of elements. This algorithm should return a pair (a, b), where a is the minimum element and b is the maximum. What is the running time of your method?

  *Hint: You can use min(a, b) and max(a, b) built-in function where each returns the minimum and the maximum number for two numbers with efficiency of $\Theta(1)$.

  1) Complete the following Pseudocode (on the next page).

Example 5 – 1)

- ✓ start and end are indices of input array A
- ✓ value_min and value_max are the minimum and maximum values

```
FIND_MIN_MAX(A, start, end) {
    if start == end
        // Case: array A has only one element.
        [                    ]
    else if end – start == 1
        [                    ]
    else
        // Divide step
        [                    ]
        // Conquer step
        [                    ]
        // Combine step code
        [                    ]
        return (value_min, value_max)
}
```

Example 5 – 2)

- What is the complexity of FIND_MIN_MAX algorithm? (Use asymptotic notation Θ)

Example 6

- Use mathematical induction to show that the solution of the recurrence

$$T(n) = \begin{cases} 2 & if\ n = 2, \\ 2T\left(\frac{n}{2}\right) + n & if\ n = 2^k, for\ k > 1 \end{cases}$$

is $T(n) = n\lg(n)$.

*Hint: Assume that n is an exact power of 2, i.e., $n = 2^k$

Example 7

- ❑ Solve the following recurrence equation by applying master theorem.

$$T(n) = 3T\left(\frac{n}{2}\right) + n^2$$

Example 8

- Solve the following recurrence equation by applying master theorem.

$$T(n) = 2^n \times T\left(\frac{n}{2}\right) + n^n$$

Example 9

- ❑ Solve the following recurrence equation by applying master theorem.

$$T(n) = \sqrt{2} \times T\left(\frac{n}{2}\right) + \log(n)$$

Example 10

- ❑ Solve the following recurrence equation by <u>applying master theorem</u>.

$$T(n) = 0.5 \times T\left(\frac{n}{2}\right) + \frac{1}{n}$$

Example 11

- Solve the following recurrence equation by applying master theorem.

$$T(n) = 3 \times T\left(\frac{n}{3}\right) + \frac{n}{2}$$